ENHANCING 3D SAR IMAGING: A NEAR-FIELD BACK PROJECTION ALGORITHM FOR ADDRESSING GEOMETRIC DISTORTION

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ABSTRACT

This paper aims to address the issue of geometric distortion in traditional 2D SAR imaging under near-field conditions. To achieve this, the study explores the geometric model of MIMO SAR 3D imaging and derives the 3D back projection (BP) imaging algorithm based on it. The algorithm's effectiveness is verified through the simulation of point targets and the collection of data using a sliding rail SAR system to image metal sheets in actual scenes. The simulation and actual imaging results confirm that our near-field BP algorithm has better imaging results compared to the BP algorithm which does not consider near-field effects while maintaining acceptable computational complexity.

Index Terms—Near-field, MIMO SAR, BP, 3D imaging

1. INTRODUCTION

3D near-field imaging has become a significant issue in SAR imaging in recent years, drawing lots of attention [1-4] . In the near field, the plane wave assumption of the radar signal is not accurate, so a more precise spherical electromagnetic wave model is required. In comparison to the low band far field, the observation angle changes significantly during each measurement in the high wave band near field, which distorts the two-dimensional image of distance and orientation. Additionally, the high-frequency influence is also significant. These factors combined make signal processing more difficult, and traditional processing methods cannot be directly applied.

MIMO SAR imaging, considering near-field effects, has become a popular research area among scholars and scientists. David M. Sheen et al. [5] have developed a wideband millimeter wave holographic surveillance system that uses coherent wideband millimeter wave illumination to create three-dimensional full-focus images for personnel monitoring. Zhuge [6] has designed a sparse MIMO SAR UWB imaging system that demonstrates high-resolution three-dimensional volumetric imaging by combining a one-dimensional MIMO array with digital beam focusing and orthogonal SAR. However, its special MIMO array requirements are often hard to meet. Yanik et al. [7] have suggested a near-field millimeter wave imaging system using two-dimensional MIMO arrays. High-quality images with fewer antenna elements can be produced [8]. However, its processing flow is complicated, making it difficult to apply quickly. In [9], an effective frequent-domain imaging algorithm for MIMO-SAR is proposed, which compensates and interpolates in the wave-number domain while converting 4-dimensional space to 3-dimensional space by accumulation. Though it achieves high-precision imaging, it is not suitable for general MIMO arrays and may have grating lobe problems. Literature [10] proposes a MIMO distance migration algorithm based on non-uniform FFT (NUFFT), called NUFFT-based MIMO-RMA, for efficient microwave imaging. But it may lead to strong artifacts in the focused image. These algorithms are frequency-domain algorithms, but they have limitations if high-resolution imaging requires high-precision motion compensation processing.

A time-domain imaging algorithm can be used to address the issues mentioned above. One algorithm is BP algorithm, which was introduced in the computer field. Theoretically, this algorithm is suitable for any orbital model and imaging mode. However, the BP algorithm has the drawback of being time-consuming and requiring high CPU requirements. To address these issues, researchers have suggested a multistage projection method to speed up the calculation. Two representative algorithms of this method are the Fast Back Projection (FBP) algorithm proposed by Ali F. in 1999 [11], and the Fast Factorized Back Projection (FFBP) algorithm proposed by Lars M.H. [12]. With the recent advancements in FPGA operation level and computer processing technology, the operation time of the BP algorithm has significantly decreased, leading to its widespread application.

This paper introduces a new imaging algorithm based on MIMO SAR. High-quality imaging results require the implementation of a cascaded radar system. However, it is challenging to eliminate phase errors between cascaded chips, which can impact imaging outcomes [13]. Therefore, this paper validates the proposed method on the state-of-the-art TI cascaded millimeter-wave radar. This paper employs the real positions of the transmitting and receiving array for data processing, rather than the virtual array positions, thereby further optimizing the algorithm's performance. With this algorithm, it is possible to efficiently generate 3D images of scenes, accurately accounting for the effects of the near-field.

2. NEAR-FIELD 3D BP IMAGING

A. Near-field MIMO SAR Signal Model

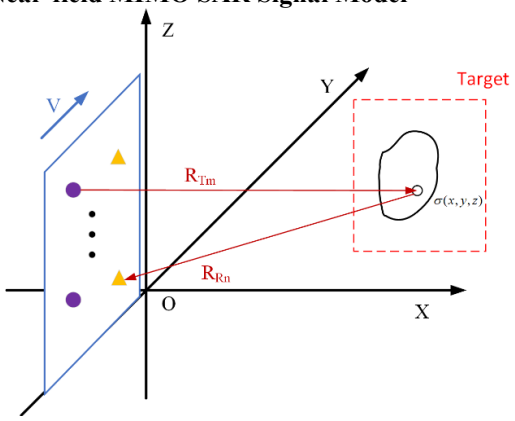


Fig. 1: Near-field MIMO SAR imaging geometric model.

The geometric model of near-field MIMO SAR imaging is shown in Fig. 1. In the Cartesian coordinate system, the x, y, and z axes represent range, azimuth, and height, respectively; a radar consisting of M transmitting elements and N receiving elements is located in the YOZ plane and scanned along the Y axis; The position of the transmitting element is represented by (x_m, y_m, z_m) and the position of the receiving element is represented by (x_n, y_n, z_n) ; the position of any point P in the target region is (x, y, z) and the reflectance is $\sigma(x, y, z)$; R_{T_m} represents the distance between the transmitting element and the target P, R_{R_n} represents the distance between the receiving element and the target P, and R_{T_m} and R_{R_n} are represented by the following formula:

$$\begin{aligned} R_{Tm} &= \sqrt{(x-x_m)^2+(y-y_m)^2+(z-z_m)^2} \\ R_{Rn} &= \sqrt{(x-x_n)^2+(y-y_n)^2+(z-z_n)^2} \end{aligned} \tag{1}$$

Let c represent the speed of light, the propagation delay between transmitting and receiving antennas can be expressed as:

$$\tau_{m,n} = \frac{R_{Tm}+R_{Rn}}{c} \tag{2}$$

Assuming that the transmitted signal is FMCW, the corresponding received IF signal is:

$$\begin{aligned} s_{m,n}(t) = &\iiint_{\Pi(x,y,z)} \sigma(x,y,z)\exp(-j2\pi f\tau_{m,n}) \\ &\exp(-j2\pi Kt\tau_{m,n})\exp(\pi K\tau_{m,n}^2)dxdydz \end{aligned} \tag{3}$$

Where $\Pi(x, y, z)$ represents the distribution space of the target, f represents the FMCW frequency, K represents the frequency modulation slope, and t represents the time.

The range Fourier transform is applied to (3) to obtain a one-dimensional range image:

$$\begin{aligned} S_{m,n} = &\iiint_{\Pi(x,y,z)} \sigma(x,y,z)\exp(-j2\pi f\tau_{m,n}) \\ &\exp(\pi K\tau_{m,n}^2)dxdydz \end{aligned} \tag{4}$$

B. Proposed Near-field Imaging Algorithm

The imaging region containing the target distribution region $\Pi(x, y, z)$ is divided into a series of three-dimensional grids. For a certain grid point (x', y', z') in the imaging region, the distance between the transceiver antenna and the grid point is obtained by geometric relation:

$$\begin{aligned} R'_{Tm} &= \sqrt{(x'-x_m)^2+(y'-y_m)^2+(z'-z_m)^2} \\ R'_{Rn} &= \sqrt{(x'-x_n)^2+(y'-y_n)^2+(z'-z_n)^2} \end{aligned} \tag{5}$$

The corresponding distance propagation delay of grid points is obtained:

$$\tau'_{m,n} = \frac{R'_{Tm}+R'_{Rn}}{c} \tag{6}$$

Interpolation of the one-dimensional distance image is necessary to compensate for the mismatch between grid points. The phase term of the grid point to be compensated can be determined based on propagation delay.

$$\varphi(\tau'_{m,n}) = \exp(j2\pi f\tau'_{m,n})\exp(-j\pi K\tau'^2_{m,n}) \tag{7}$$

The value of the grid point after phase compensation is denoted as $I(x', y', z')$, then:

$$I(x', y', z') = S_{m,n} \bullet \exp(j2\pi f\tau'_{m,n})\exp(-j\pi K\tau'^2_{m,n}) \tag{8}$$

Traverse all antennas and azimuths, and coherently superposition the back-projection values of the corresponding grid points.

$$A(x', y', z') = \sum_{k=1}^{N_k}\sum_{M}\sum_{N} I(x', y', z') \tag{9}$$

where N_k represents the number of azimuth points, and $A(x', y', z')$ represents the back projection value of the grid point after coherent superposition. It can be found that when the distance between the grid point and the radar is exactly equal to the distance between the target and the radar, the amplitude of the back projection value is higher, while the back projection value of the non-target region is smaller.

It is not difficult to find that in the target region $A(x', y', z') \approx \sigma(x, y, z)$, the image of the target region can be realized by this algorithm.

The flow chart for the near-field 3D BP algorithm is presented in Fig. 2. The process can be broken down into four distinct steps.

Step 1: Range FFT. The range FFT of the received IF signal is carried out to complete the range compression. At the same time, to match the range of each grid point with the one-dimensional range image as much as possible, it is necessary to interpolate the one-dimensional range image.

Step 2: Imaging area division and phase compensation. The imaging area of interest is divided into grids of appropriate size, and the distance between each grid point and the radar is calculated by (5). Then, the phase compensation is realized by (6)-(8), and the back projection value is obtained.

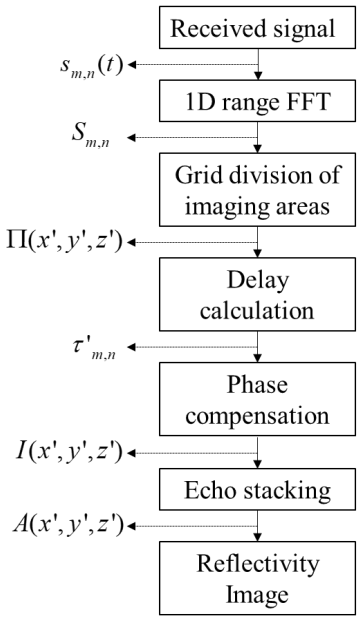


Fig. 2: 3D BP flow chart.

Note that the radar transceiver mode in this paper is MIMO. In general, when calculating the distance between the grid point and the radar, the default position of the radar transceiver array is the virtual array position caused by MIMO. However, in near-field conditions, the use of virtual element position will result in distortion of imaging results. Therefore, when calculating the range in step 2, the actual position of the radar transceiver array should be used.

Step 3: Traverse all azimuth directions to achieve coherent superposition of the back projection values. For the target region, the backward projected value presents a larger amplitude in all directions, while for the non-target region, the backward projected value presents a larger value in only a few directions. Therefore, after the coherent stack is completed by (9), the target region is represented as a larger value and the non-target region appears as noise.

Step 4: Image output.

3. SIMULATION AND RESULTS

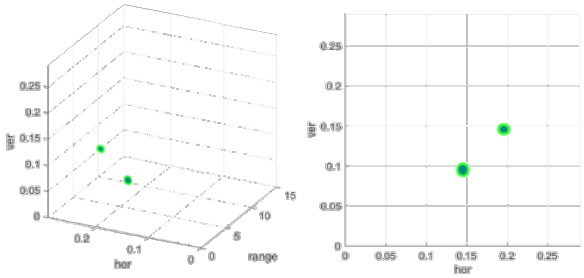
A. Algorithm Simulation

In this section, we utilize a radar system that operates at 77 GHz millimeter wave frequency to simulate the algorithm. The radar platform employs MIMO technology to virtualize 86 array elements in the height dimension. Additionally, it enhances resolution through synthetic aperture in the azimuth direction. To align with the parameters of the subsequent actual imaging system, we have selected simulation parameters which are listed in Table I.

We establish two targets in simulated imaging scene, with coordinates of $(0.15m, 0.6m, 0.1m)$ and $(0.1m, 0.7m, 0.15m)$, respectively. Create the radar echo signal. Fig. 3 displays the results following processing by the near-field 3D BP imaging technique.

TABLE I Main simulation parameters

Parameter	Value
Bandwidth B (GHz)	3.20
Frequency f_0 (GHz)	77
SAR Platform speed v (m/s)	0.02
Chirp rate K (MHz/us)	100
Synthetic aperture length L (m)	0.66
Range resolution Res_r (cm)	5.93
Azimuth resolution Res_{az} (cm)	0.65
Height resolution Res_h (cm)	2.32
Range sampling points N_s	256
Sampling rate f_s (MHz)	8
Azimuth frame N_f	330
Frame duration T_f (ms)	100



(a) 3D image. (b) Range slice.

Fig. 3: Point target simulation image.

Simulation results confirm the efficacy of the suggested near-field 3D BP algorithm. The imaging center locations of the two points match the predetermined target position.

B. Imaging Results

To test the proposed near-field BP imaging algorithm, we have built a sliding-rail SAR 3D imaging system. The system comprises a personal computer (PC), GVM electric control rails, and a TI millimeter-wave radar with the plate number AWR2243. The frame diagram for the radar system is depicted in Fig. 4.

Three hardware modules from Texas Instruments compose the mmWave radar board: the acquisition board, the MMWave-DevPack, and the Boost. Fig. 5 displays the configuration of radar antennas.

For the real scenario, an iron target in the shape of the letter "A" was placed one meter away from the radar Doppler center at an angle. The radar was configured according to Table I. Fig. 6(a) displays the dimensions of the target, while Fig. 6(b) shows the target scene.

The near-field 3D BP algorithm is used to concentrate the target signal after IF signal from the radar echo is collected by the MIMO SAR three-dimensional imaging system. Fig. 7 illustrates how to determine the target's location distribution by choosing a suitable intensity threshold.

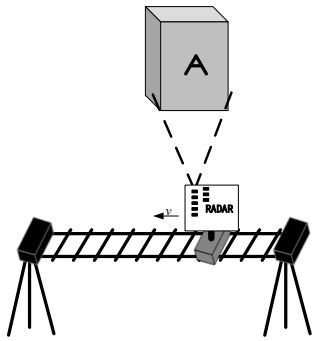


Fig. 4: MIMO SAR 3D imaging system.

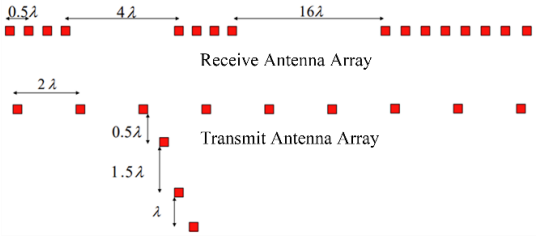
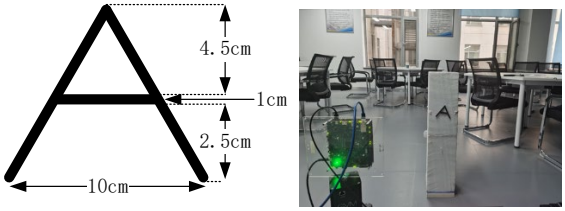


Fig. 5: MIMO array topology.



(a) "A" iron sheet size. (b) Imaging scene.

Fig. 6: Target imaging scene.

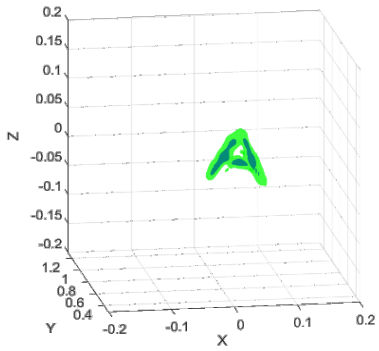


Fig. 7: Target 3D image.

Fig. 8 displays the imaging results obtained from the near-field 3D BP technique, whereby the imaging slice at 1m in the 3D imaging signal is chosen. The image result displays the letter A, and the size is in line with the model, as the figure illustrates. On the other hand, Fig. 9 displays the imaging results of the 3D BP method without taking the near-field condition into account. It is not possible to focus due to the significant distortion issue in the height dimension. The proposed near-field BP imaging technique is proven to be effective by the experimental results.

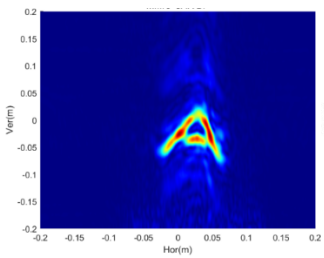


Fig. 8: Near-field result.

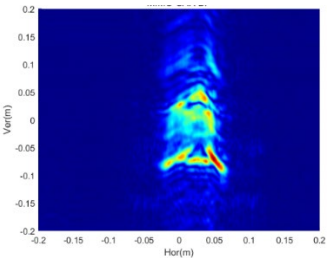


Fig. 9: Non-near-field result.

4. CONCLUSION

This work proposes a near-field 3D BP imaging algorithm to address the distortion issue of near-field SAR imaging. The simulation of numerous targets and the real measurement using the sliding-rail SAR system validate the imaging algorithm's efficacy. When compared to the conventional BP imaging method, the near-field condition-aware BP algorithm is able to perform high-precision focus imaging of the target at a closer range with more accuracy. The suggested imaging method is capable of performing high-resolution imaging in actual settings, according to simulation and experimental results.

5. REFERENCES

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